

Adv DevOps CaseStudy

Kubernetes Application with Basic Monitoring Concepts Used: Kubernetes, AWS Cloud9, and Nagios.

Problem Statement: "Deploy a simple application (e.g., an Nginx server) on a Kubernetes cluster using AWS Cloud9 and monitor its health using Nagios."

Tasks: Use AWS Cloud9 to deploy the Nginx server on a Kubernetes cluster. Install and configure Nagios to monitor the Nginx pod's availability. Verify that Nagios can detect when the Nginx pod is running or not.

Theory Behind Deploying Applications on Kubernetes and Monitoring with Nagios

1. Kubernetes Overview

Kubernetes is an open-source container orchestration platform designed to automate deploying, scaling, and managing containerized applications. Its key components include:

- **Pods:** The smallest deployable units in Kubernetes, which can hold one or more containers.
- **Services:** Abstracts the access to a set of pods, enabling communication and load balancing.
- **Deployments:** Manage the lifecycle of pods, ensuring the desired state is maintained.

Kubernetes facilitates high availability, scalability, and efficient resource utilization of applications in a distributed environment .

2. AWS Cloud9

AWS Cloud9 is a cloud-based integrated development environment (IDE) that allows developers to write, run, and debug code with just a browser. It provides:

- **Preconfigured Environments:** Easily set up for various development tasks.
- **Collaboration Features:** Multiple developers can work on the same project simultaneously.
- **Integrated Terminal:** Direct access to AWS services and the ability to run shell commands .

Using AWS Cloud9 to deploy applications on Kubernetes simplifies the process by providing a unified interface for coding, testing, and deploying.

3. Nginx Server

Nginx is a high-performance web server that can also be used as a reverse proxy, load balancer, and HTTP cache. Deploying Nginx in a Kubernetes cluster is common for serving static content and handling web traffic efficiently. Its lightweight nature makes it suitable for containerized environments .

4. Monitoring with Nagios

Nagios is an open-source monitoring system that enables users to monitor systems, networks, and infrastructure. Key features include:

- **Real-Time Monitoring:** Tracks the status of hosts and services in real-time.
- **Alerting:** Sends notifications when issues arise, helping to address problems proactively.
- **Plugins:** Extensible through a wide range of plugins, allowing customized monitoring for various services, including Nginx .

For this project, Nagios will monitor the availability and health of the Nginx pod, ensuring it is running as expected.

Deployment and Monitoring Workflow

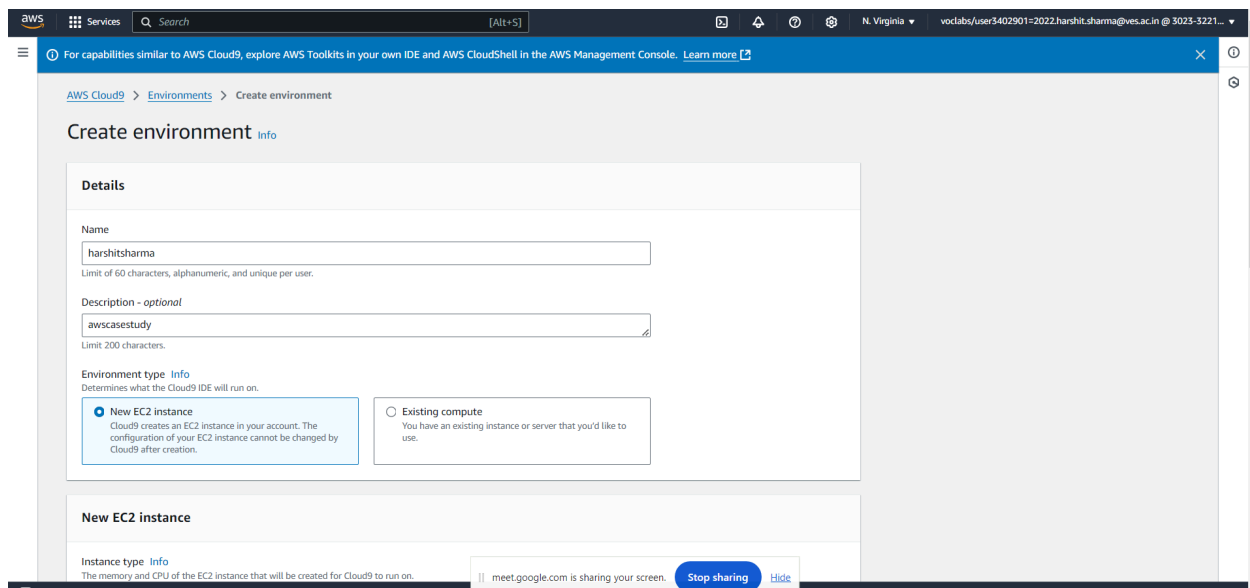
1. Deploying Nginx on Kubernetes:

- **Use AWS Cloud9 to set up a Kubernetes environment.**

- Create a deployment for the Nginx server, exposing it via a service for external access.
- 2. Configuring Nagios:**
- Install Nagios on a server or virtual machine that has access to the Kubernetes cluster.
 - Define commands in Nagios to check the Nginx pod's status, using Kubernetes commands (like `kubectl get pods`).
 - Set up alerts to notify administrators if the Nginx pod goes down.
- 3. Verifying Monitoring:**
- Check that Nagios correctly identifies the state of the Nginx pod and sends alerts as expected.
 - Test the setup by stopping the Nginx pod and ensuring Nagios detects the change.

Steps:

1)Create Environment in cloud9



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The image shows two screenshots from the AWS Management Console. The top screenshot is the 'New EC2 instance' wizard. It displays three instance type options: t2.micro (1 GiB RAM + 1 vCPU), t3.small (2 GiB RAM + 2 vCPU), and m5.large (8 GiB RAM + 2 vCPU). The t3.small instance is selected. Below the instance types, there are options for 'Additional instance types', 'Platform' (set to Amazon Linux 2023), and 'Timeout' (set to Never). The 'Network settings' section shows the 'Connection' method set to 'Secure Shell (SSH)'. The bottom screenshot shows the AWS Cloud9 'Environments' page. It displays a table with one environment named 'harshitsharma'. The table columns are Name, Cloud9 IDE, Environment type, Connection, Permission, and Owner ARN. The environment 'harshitsharma' is of type 'EC2 instance' and uses 'Secure Shell (SSH)' for connection. The owner ARN is 'arn:aws:sts::302332216982:assumed-role/voclabs/user3402901=2022.harshit.sharma@ves.ac.in'.

New EC2 instance

Instance type [Info](#)
The memory and CPU of the EC2 instance that will be created for Cloud9 to run on.

☐ t2.micro (1 GiB RAM + 1 vCPU)
Free-tier eligible. Ideal for educational users and exploration.

☒ t3.small (2 GiB RAM + 2 vCPU)
Recommended for small web projects.

☐ m5.large (8 GiB RAM + 2 vCPU)
Recommended for production and most general-purpose development.

☐ Additional instance types
Explore additional instances to fit your need.

Platform [Info](#)
This will be installed on your EC2 instance. We recommend Amazon Linux 2023.

Amazon Linux 2023

Timeout
How long Cloud9 can be inactive (no user input) before auto-hibernating. This helps prevent unnecessary charges.

Never

Network settings [Info](#)

Connection
How your environment is accessed.

☐ AWS Systems Manager (SSM)
Accesses environment via SSM without opening inbound ports (no ingress).

☒ Secure Shell (SSH)
Accesses environment directly via SSH, opens inbound ports.

AWS Cloud9

My environments
Shared with me
All account environments
Documentation

Successfully created **harshitsharma**. To get the most out of your environment, see [Best practices for using AWS Cloud9](#).

For capabilities similar to AWS Cloud9, explore AWS Toolkits in your own IDE and AWS CloudShell in the AWS Management Console. [Learn more](#).

AWS Cloud9 > Environments

Environments (1) [Delete](#) [View details](#) [Open in Cloud9](#) [Create environment](#)

My environments

	Name	Cloud9 IDE	Environment type	Connection	Permission	Owner ARN
☐	harshitsharma	Open	EC2 instance	Secure Shell (SSH)	Owner	arn:aws:sts::302332216982:assumed-role/voclabs/user3402901=2022.harshit.sharma@ves.ac.in

2)Open Cloud9 Terminal

The image shows the AWS Cloud9 IDE interface. The main window displays the 'Welcome to your development environment' message. Below this, there is a 'Toolkit for AWS Cloud9' section. On the right side, there are two panels: 'Getting started' and 'Configure AWS Cloud9'. The 'Getting started' panel contains links for 'Create File', 'Upload Files...', and 'Clone from GitHub'. The 'Configure AWS Cloud9' panel shows the 'Main Theme' set to 'light_dark'.

AWS Cloud9
Welcome to your development environment

AWS Cloud9 allows you to write, run, and debug your code with just a browser. You can [tour the IDE](#), [write code for AWS Lambda and Amazon API Gateway](#), [share your IDE](#) with others in real time, and much more.

Toolkit for AWS Cloud9

The AWS Toolkit for Cloud9 is an IDE extension that simplifies accessing and interacting with resources from services such as AWS Lambda, AWS CloudFormation, and AWS API Gateway. With the toolkit, developers can also develop, debug, and deploy applications using the AWS Serverless Application Model (SAM). [Learn more](#)

Getting started

[Create File](#)

[Upload Files...](#)

[Clone from GitHub](#)

Configure AWS Cloud9

Main Theme: [light_dark](#)

bash - "ip-172-31-37-0.ec" Immediate

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2.1) Install Docker

Docker is needed to run containers that will host your applications.

1. Update the Package List:

- In the terminal at the bottom of your Cloud9 IDE, type the following command and press Enter:

```
sudo yum update -y
```

The image consists of two screenshots. The top screenshot shows the AWS Management Console 'Instances' page. A table lists instances, with one instance 'aws-cloud9-ha...' in a 'Running' state. Below the table, the details for instance 'i-08ffda0fe7974cbaf' are shown, including its public IPv4 address (34.238.163.130) and public IPv4 DNS name (ec2-34-238-163-130.compute-1.amazonaws.com). The bottom screenshot shows the Cloud9 IDE interface. The terminal window displays the output of the command 'sudo yum update -y', which includes a warning about a newer release of Amazon Linux being available and a message that dependencies are resolved.

AWS Management Console - Instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
aws-cloud9-ha...	i-08ffda0fe7974cbaf	Running	t3.small	3/3 checks passed	View alarms	us-east-1a	ec2-34-238-163-130.co...

Instance details: i-08ffda0fe7974cbaf

- Instance ID: i-08ffda0fe7974cbaf
- Public IPv4 address: 34.238.163.130
- Private IPv4 addresses: 172.31.37.0
- Instance state: Running
- Public IPv4 DNS: ec2-34-238-163-130.compute-1.amazonaws.com
- Private IP DNS name (IPv4 only): ip-172-31-37-0.ec2.internal

Cloud9 IDE Terminal

```
bash - ip-172-31-37-0.ec2.x Immediate
voclabs:~/environment $ sudo yum update -y
Last metadata expiration check: 0:10:54 ago on Wed Oct 23 06:13:39 2024.

WARNING:
  A newer release of "Amazon Linux" is available.

Available Versions:
  Version 2023.6.20241010:
    Run the following command to upgrade to 2023.6.20241010:
      dnf upgrade --releasever=2023.6.20241010

Release notes:
  https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.6.20241010.html

Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $
```

2. Install Docker:

- Type the following command and press Enter:

```
sudo yum install docker -y
```

```
=====
Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $ sudo yum install docker -y
Last metadata expiration check: 0:14:02 ago on Wed Oct 23 06:13:39 2024.
Package docker-25.0.6-1.amzn2023.0.2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $ █
```

3.Start Docker:

- Run this command to start the Docker service:

```
sudo service docker start
```

```
voclabs:~/environment $ sudo service docker start
Redirecting to /bin/systemctl start docker.service
voclabs:~/environment $ █
```

4.Add Your User to the Docker Group:

- This allows you to run Docker commands without using `sudo`. Type:

bash

Copy code

```
sudo usermod -aG docker ec2-user
```

```
voclabs:~/environment $ sudo service docker start
Redirecting to /bin/systemctl start docker.service
voclabs:~/environment $ sudo usermod -aG docker ec2-user
voclabs:~/environment $ █
```

2.2 Install kubectl

kubect1 is a tool to manage Kubernetes from the command line.

1. Download the kubect1 tool:

```
curl -LO
"https://storage.googleapis.com/kubernetes-release/release/$(curl
-s
https://storage.googleapis.com/kubernetes-release/release/stable.
txt)/bin/linux/amd64/kubect1"
```

```
voclabs:~/environment $ curl -LO "https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubect1"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 53.7M 100 53.7M 0 0 89.4M 0 --:--:-- --:--:-- --:--:-- 89.6M
voclabs:~/environment $
```

2. Make it executable (like turning on a switch):

```
chmod +x ./kubect1
```

```
Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $ sudo yum install docker -y
Last metadata expiration check: 0:14:02 ago on Wed Oct 23 06:13:39 2024.
Package docker-25.0.6-1.amzn2023.0.2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $ sudo service docker start
Redirecting to /bin/systemctl start docker.service
voclabs:~/environment $ sudo usermod -aG docker ec2-user
voclabs:~/environment $ curl -LO "https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubect1"
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 53.7M 100 53.7M 0 0 89.4M 0 --:--:-- --:--:-- --:--:-- 89.6M
voclabs:~/environment $ ls -l ./kubect1
-rwxr-xr-x. 1 ec2-user ec2-user 56381592 Oct 23 06:30 ./kubect1
voclabs:~/environment $
```

```
voclabs:~/environment $ chmod +x ./kubect1
voclabs:~/environment $ ls -l ./kubect1
-rwxr-xr-x. 1 ec2-user ec2-user 56381592 Oct 16 14:30 ./kubect1
voclabs:~/environment $
```

3. Move it to your system path so you can use it:

```
sudo mv ./kubect1 /usr/local/bin/kubect1
```

```
voclabs:~/environment $ sudo mv ./kubectl /usr/local/bin/kubectl
voclabs:~/environment $ ls -l /usr/local/bin/kubectl
ls: cannot access '/usr/local/bin/kubectl': No such file or directory
voclabs:~/environment $ ls -l /usr/local/bin/kubectl
-rwxr-xr-x. 1 ec2-user ec2-user 56381592 Oct 23 06:30 /usr/local/bin/kubectl
voclabs:~/environment $ █
```

2.3 Install Minikube

Minikube lets you run Kubernetes on your own computer.

1. Download Minikube:

```
curl -Lo minikube
```

<https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64>

```
voclabs:~/environment $ curl -Lo minikubehttps://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
curl: (2) no URL specified
curl: try 'curl --help' for more information
voclabs:~/environment $ curl -Lo minikube https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 99.0M  100 99.0M    0     0  78.5M      0  0:00:01  0:00:01 --:--:-- 78.6M
voclabs:~/environment $ █
```

2.Make the Minikube Binary Executable: Run the command below to give permission to the downloaded file to be executed:

```
chmod +x minikube
```

```
voclabs:~/environment $ chmod +x minikube
voclabs:~/environment $ ls -l ./minikube
-rwxr-xr-x. 1 ec2-user ec2-user 103820392 Oct 23 06:39 ./minikube
voclabs:~/environment $ █
```

3.Move Minikube to a Directory in Your PATH: Move the Minikube binary to `/usr/local/bin`, which is a common directory for executable files:

```
sudo mv minikube /usr/local/bin/
```

```
voclabs:~/environment $ sudo mv minikube /usr/local/bin/
voclabs:~/environment $ ls -l /usr/local/bin/
total 156456
lrwxrwxrwx. 1 root    root          37 Oct  8 02:51 aws -> /usr/local/aws-cli/v2/current/bin/aws
lrwxrwxrwx. 1 root    root          47 Oct  8 02:51 aws_completer -> /usr/local/aws-cli/v2/current/bin/aws_completer
lrwxrwxrwx. 1 root    root          31 Oct  8 03:02 cdk -> ../lib/node_modules/cdk/bin/cdk
-rwxr-xr-x. 1 root    root        1025 Oct  8 03:03 git-remote-codecommit
-rwxr-xr-x. 1 ec2-user ec2-user 56381592 Oct 23 06:30 kubectl
-rwxr-xr-x. 1 ec2-user ec2-user 103820392 Oct 23 06:39 minikube
lrwxrwxrwx. 1 root    root          38 Oct  8 02:54 npm -> ../lib/node_modules/npm/bin/npm-cli.js
lrwxrwxrwx. 1 root    root          38 Oct  8 02:54 npx -> ../lib/node_modules/npm/bin/npx-cli.js
lrwxrwxrwx. 1 root    root          45 Oct  8 03:02 prettier -> ../lib/node_modules/prettier/bin/prettier.cjs
lrwxrwxrwx. 1 root    root          38 Oct  8 03:01 sam -> /usr/local/aws-sam-cli/current/bin/sam
voclabs:~/environment $ █
```

2.4 Start Minikube

1. Run Minikube (this starts Kubernetes):

```
minikube start --driver=docker
```

```
voclabs:~/environment $ minikube start --driver=docker
* minikube v1.34.0 on Amazon 2023.5.20241001
* Using the docker driver based on user configuration

X The requested memory allocation of 1901MiB does not leave room for system overhead (total system memory: 1901MiB). You may face stability issues.
* Suggestion: Start minikube with less memory allocated: 'minikube start --memory=1901mb'

* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.45 ...
* Downloading Kubernetes v1.31.0 preload ...
  > preloaded-images-k8s-v18-v1...: 326.69 MiB / 326.69 MiB 100.00% 66.50 M
  > gcr.io/k8s-minikube/kicbase...: 487.90 MiB / 487.90 MiB 100.00% 60.55 M
^C
voclabs:~/environment $ █
```

2. Check if Kubernetes is running by typing:

```
kubectl get nodes
```

```
voclabs:~/environment $ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

voclabs:~/environment $ kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
minikube      Ready     control-plane   47s   v1.31.0
voclabs:~/environment $ █
```

3) Deploy Nginx on Kubernetes

Nginx is a web server. Now, we'll deploy it on Kubernetes.

3.1 Create Nginx Deployment

We tell Kubernetes to create a pod (a small unit that runs our app).

1. Run this command to deploy Nginx:

```
kubectl create deployment nginx --image=nginx
```

```
minikube   Ready   control-plane   4/5   v1.31.0
voclabs:~/environment $ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
voclabs:~/environment $ █
```

3.2 Expose Nginx

We need to expose the Nginx server so we can access it from a browser.

1. Run this command to expose Nginx on port 80:

```
kubectl expose deployment nginx --type=NodePort --port=80
```

```
voclabs:~/environment $ kubectl expose deployment nginx --type=NodePort --port=80
service/nginx exposed
voclabs:~/environment $ █
```

3.3 Get the Service URL

1. To access the Nginx server, run this command:

```
minikube service nginx
```

It will show a URL. Copy that URL and open it in a browser. You should see the Nginx welcome page!

```
bash - "ip-172-31-37-0.ec x Immediate x (+)
```

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime

* Related issue: <https://github.com/kubernetes/minikube/issues/9024>

* Preparing Kubernetes v1.31.0 on Docker 27.2.0 ...

- Generating certificates and keys ...
- Booting up control plane ...
- Configuring RBAC rules ...

* Configuring bridge CNI (Container Networking Interface) ...

* Verifying Kubernetes components...

- Using image gcr.io/k8s-minikube/storage-provisioner:v5

* Enabled addons: default-storageclass, storage-provisioner

* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

```
voclabs:~/environment $ ^C
voclabs:~/environment $ minikube status
minikube
type: Control Plane
host: Running
kublet: Running
apiserver: Running
kubeconfig: Configured

voclabs:~/environment $ kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
minikube      Ready    control-plane   47s   v1.31.0

voclabs:~/environment $ kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
voclabs:~/environment $ kubectl expose deployment nginx --type=NodePort --port=80
service/nginx exposed
voclabs:~/environment $ minikube service nginx
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx	80	http://192.168.49.2:30982

```

* Opening service default/nginx in default browser...
http://192.168.49.2:30982
voclabs:~/environment $ █
```

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```
voclabs:~/environment $ kubectl expose deployment nginx --type=NodePort --port=80
service/nginx exposed
voclabs:~/environment $ minikube service nginx
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx	80	http://192.168.49.2:30982

```
* Opening service default/nginx in default browser...
http://192.168.49.2:30982
voclabs:~/environment $ █
```

As Nginx server is not opening

Step to Exec into the Pod

Run the Exec Command: Use the following command to exec into the pod:

```
kubectl exec -it nginx-676b6c5bbc-6x2q7 -- /bin/bash
```

Test Connectivity: Once you're inside the pod, run:

```
curl localhost:80
```

```
http://192.168.49.2:30982
voclabs:~/environment $ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-676b6c5bbc-b77k7             1/1     Running   0           5m55s
voclabs:~/environment $ kubectl exec -it nginx-676b6c5bbc-b77k7 -- /bin/bash
root@nginx-676b6c5bbc-b77k7:/# curl localhost:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@nginx-676b6c5bbc-b77k7:/# █
```

Using Curl in the Terminal (to Test)

If you want to see the output directly in the terminal, you can run:

```
curl http://192.168.49.2:31320
```

```
voclabs:~/environment $ curl http://192.168.49.2:30982
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
voclabs:~/environment $ █
```

4) Install and Configure Nagios

1.Install Nagios

1. SSH into your Cloud9 environment (if not already):

ssh -i "id_rsa" root@ec2-3-94-99-227.compute-1.amazonaws.com

```
voclabs:~/environment $ kubectl get services
NAME         TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes   ClusterIP   10.96.0.1     <none>         443/TCP          14m
nginx        NodePort    10.100.145.92 <none>         80:30982/TCP     12m
voclabs:~/environment $ http://192.168.49.2:30982
bash: http://192.168.49.2:30982: No such file or directory
voclabs:~/environment $ curl http://192.168.49.2:30982
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
voclabs:~/environment $ ssh -i "id_rsa" root@ec2-34-238-163-130.compute-1.amazonaws.com
Warning: Identity file id_rsa not accessible: No such file or directory.
The authenticity of host 'ec2-34-238-163-130.compute-1.amazonaws.com (172.31.37.0)' can't be established.
ED25519 key fingerprint is SHA256:coNNeP6pL5t5USq7IICUIhMLCr/n5Au9Uh4JWGNJBuG.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-238-163-130.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
root@ec2-34-238-163-130.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
voclabs:~/environment $ █
```

2.Update the package manager:

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`sudo yum update -y # For Amazon Linux`

```
voclabs:~/environment $ ssh -i "id_rsa" root@ec2-34-238-163-130.compute-1.amazonaws.com
Warning: Identity file id_rsa not accessible: No such file or directory.
The authenticity of host 'ec2-34-238-163-130.compute-1.amazonaws.com (172.31.37.0)' can't be established.
ED25519 key fingerprint is SHA256:coNNeP6pL5t5USq7IICUIhMLCr/n5Au9Uh4JWGNJBUG.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-238-163-130.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
root@ec2-34-238-163-130.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
voclabs:~/environment $ sudo yum update -y # For Amazon Linux
Last metadata expiration check: 0:57:10 ago on Wed Oct 23 06:13:39 2024.
```

=====

WARNING:

A newer release of "Amazon Linux" is available.

Available Versions:

Version 2023.6.20241010:

Run the following command to upgrade to 2023.6.20241010:

```
dnf upgrade --releasever=2023.6.20241010
```

Release notes:

<https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.6.20241010.html>

=====

Dependencies resolved.

Nothing to do.

Complete!

voclabs:~/environment \$ █

Install required dependencies:

`sudo yum install -y gcc glibc glibc-common make autoconf httpd php gcc-c++ perl`

`sudo yum install -y gd-devel make automake`

```
voclabs:~/environment $ sudo yum install -y gcc glibc glibc-common make autoconf httpd php gcc-c++ perl
sudo yum install -y gd-devel make automake
Last metadata expiration check: 0:57:58 ago on Wed Oct 23 06:13:39 2024.
Package gcc-11.4.1-2.amzn2023.0.2.x86_64 is already installed.
Package glibc-2.34-52.amzn2023.0.11.x86_64 is already installed.
Package glibc-common-2.34-52.amzn2023.0.11.x86_64 is already installed.
Package make-1:4.3-5.amzn2023.0.2.x86_64 is already installed.
Package autoconf-2.69-36.amzn2023.0.3.noarch is already installed.
Package httpd-2.4.62-1.amzn2023.x86_64 is already installed.
Package php8.2-8.2.21-1.amzn2023.0.1.x86_64 is already installed.
Package gcc-c++-11.4.1-2.amzn2023.0.2.x86_64 is already installed.
Package perl-4:5.32.1-477.amzn2023.0.6.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
voclabs:~/environment $ █
```

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Download and Install Nagios 4.4.9

1. Change to the `/tmp` directory:

```
cd /tmp
```

```
voclabs:~/environment $ cd /tmp
```

Download Nagios 4.4.9:

```
wget
```

```
https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.4.9.tar.gz
```

```
voclabs:~/environment $ cd /tmp
voclabs:/tmp $ wget https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.4.9.tar.gz
--2024-10-23 07:12:43-- https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.4.9.tar.gz
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/NagiosEnterprises/nagioscore/tar.gz/refs/tags/nagios-4.4.9 [following]
--2024-10-23 07:12:44-- https://codeload.github.com/NagiosEnterprises/nagioscore/tar.gz/refs/tags/nagios-4.4.9
Resolving codeload.github.com (codeload.github.com)... 140.82.113.10
Connecting to codeload.github.com (codeload.github.com)|140.82.113.10|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/x-gzip]
Saving to: 'nagios-4.4.9.tar.gz'

nagios-4.4.9.tar.gz                                [   <=>]

2024-10-23 07:12:45 (17.3 MB/s) - 'nagios-4.4.9.tar.gz' saved [11339485]

voclabs:/tmp $ █
```

Extract the downloaded file:

```
tar -zxvf nagios-4.4.9.tar.gz
```

```
nagioscore-nagios-4.4.9/tap/tests/plan/too-many-tests/Makefile.in
nagioscore-nagios-4.4.9/tap/tests/plan/too-many-tests/test.c
nagioscore-nagios-4.4.9/tap/tests/plan/too-many-tests/test.pl
nagioscore-nagios-4.4.9/tap/tests/plan/too-many-tests/test.t
nagioscore-nagios-4.4.9/tap/tests/skip/
nagioscore-nagios-4.4.9/tap/tests/skip/Makefile.am
nagioscore-nagios-4.4.9/tap/tests/skip/Makefile.in
nagioscore-nagios-4.4.9/tap/tests/skip/test.c
nagioscore-nagios-4.4.9/tap/tests/skip/test.pl
nagioscore-nagios-4.4.9/tap/tests/skip/test.t
nagioscore-nagios-4.4.9/tap/tests/todo/
nagioscore-nagios-4.4.9/tap/tests/todo/Makefile.am
nagioscore-nagios-4.4.9/tap/tests/todo/Makefile.in
nagioscore-nagios-4.4.9/tap/tests/todo/test.c
nagioscore-nagios-4.4.9/tap/tests/todo/test.pl
nagioscore-nagios-4.4.9/tap/tests/todo/test.t
nagioscore-nagios-4.4.9/test/
nagioscore-nagios-4.4.9/test/test-downtime.pl
nagioscore-nagios-4.4.9/update-version
nagioscore-nagios-4.4.9/worker/
nagioscore-nagios-4.4.9/worker/Makefile.in
nagioscore-nagios-4.4.9/worker/ping/
nagioscore-nagios-4.4.9/worker/ping/.gitignore
nagioscore-nagios-4.4.9/worker/ping/Makefile.in
nagioscore-nagios-4.4.9/worker/ping/worker-ping.c
nagioscore-nagios-4.4.9/xdata/
nagioscore-nagios-4.4.9/xdata/.gitignore
nagioscore-nagios-4.4.9/xdata/Makefile.in
nagioscore-nagios-4.4.9/xdata/xcddefault.c
nagioscore-nagios-4.4.9/xdata/xcddefault.h
nagioscore-nagios-4.4.9/xdata/xodtemplate.c
nagioscore-nagios-4.4.9/xdata/xodtemplate.h
nagioscore-nagios-4.4.9/xdata/xpddefault.c
nagioscore-nagios-4.4.9/xdata/xpddefault.h
nagioscore-nagios-4.4.9/xdata/xrddefault.c
nagioscore-nagios-4.4.9/xdata/xrddefault.h
nagioscore-nagios-4.4.9/xdata/xsddefault.c
nagioscore-nagios-4.4.9/xdata/xsddefault.h
```

Navigate to the extracted Nagios directory:

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cd /tmp/nagioscore-nagios-4.4.9

```
voclabs:/tmp $ ls /tmp
init-yum-update-security      juju-mk9a8c725ee9c1b19a2220eb76ac32d84e5f9db3  nagios-4.4.9.tar.gz
juju-mk8c85d4d088b7cdfc65fc61ffc33c56f44f92e6  juju-mk9bab7919fc13e4b960d13c91e47d3fae6c575e  nagios-4.4.9.tar.gz.1
juju-mk133e77b266db33c494116c3054e2c2f1375135  juju-mk808f59e510b0d424ad13a91ccb2dacf7c7277f  nagioscore-nagios-4.4.9
juju-mk1d30f2bcdedcd5224458206a60e0bb002d6810  juju-mkbb355815a506add29c5d0b6b99541b24ddfb09  systemd-private-9973b884c5e547ed8dfa4486860535e8-chronyd.service-jXBdLZ
juju-mk23eace0344495582ff3ffc34ef64c5e2823c0b  juju-mkd0264cab96b8158ea147e8e92a5ca2fc201f4b  systemd-private-9973b884c5e547ed8dfa4486860535e8-dbus-broker.service-VQDhSM
juju-mk31d488c08ab9e96d36b37d93edbcda9c61de3f  juju-mke39e9c84c97a4cc79f46eddd6372cb206cf268  systemd-private-9973b884c5e547ed8dfa4486860535e8-policy-routes@ens5.service-VPpUln
juju-mk48b97e177eed8b45b2497b4a5355eb8b18afbe  juju-mkef9aaff15d283d6b97d136c6699af16b09f318  systemd-private-9973b884c5e547ed8dfa4486860535e8-systemd-logind.service-va0FKG
juju-mk5f377c42c55cd70f48bdad1ac96608178ba470  juju-mkf0813785b02e8c2f81710723758be1ce3495c7  systemd-private-9973b884c5e547ed8dfa4486860535e8-systemd-resolved.service-JmPw5Z
juju-mk6371d46da52852a18e5d36c47a6130f62459d0  minikube_ip_f509b92a6bf47a50fbcf69b9f742b8f50c84fde6_0.log  tmux-1000
juju-mk69bd8dc7d2764264cc5a3f0a64dd12f700bbe1  minikube_service_292a3760020cf35ab6982964f5797e02f86fc8f6_0.log
juju-mk88f1e613813ac73655cb16d65fe871e5b8acbf  minikube_status_85d757c40b021391bff41828f62cc0171dd27151_0.log
voclabs:/tmp $ cd /tmp/nagioscore-nagios-4.4.9
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Compile and Install Nagios:

./configure --with-command-group=nagios

c

sudo make install

sudo make install-init

sudo make install-config

sudo make install-commandmode

General Options:

```
-----  
Nagios executable: nagios  
Nagios user/group: nagios,nagios  
Command user/group: nagios,nagios  
Event Broker: yes  
Install ${prefix}: /usr/local/nagios  
Install ${includedir}: /usr/local/nagios/include/nagios  
Lock file: /run/nagios.lock  
Check result directory: /usr/local/nagios/var/spool/checkresults  
Init directory: /lib/systemd/system  
Apache conf.d directory: /etc/httpd/conf.d  
Mail program: /bin/mail  
Host OS: linux-gnu  
IOBroker Method: epoll
```

Web Interface Options:

```
-----  
HTML URL: http://localhost/nagios/  
CGI URL: http://localhost/nagios/cgi-bin/  
Traceroute (used by WAP): /usr/bin/traceroute
```

Review the options above for accuracy. If they look okay,
type 'make all' to compile the main program and CGIs.

*** Support Notes *****

If you have questions about configuring or running Nagios, please make sure that you:

- Look at the sample config files
- Read the documentation on the Nagios Library at:
<https://library.nagios.com>

before you post a question to one of the mailing lists. Also make sure to include pertinent information that could help others help you. This might include:

- What version of Nagios you are using
- What version of the plugins you are using
- Relevant snippets from your config files
- Relevant error messages from the Nagios log file

For more information on obtaining support for Nagios, visit:

<https://support.nagios.com>

Enjoy.

voclabs:/tmp/nagioscore-nagios-4.4.9 \$ █

Set Up Nagios Web Interface:

```
sudo make install-webconf
```

```
sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-webconf
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/httpd/conf.d/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/httpd/conf.d/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Creating Nagios User

Run the following command to create the **nagios** user and group:

```
sudo useradd nagios
```

```
sudo groupadd nagios
```

```
voclabs:~ $ sudo useradd nagios
sudo groupadd nagios
voclabs:~ $ getent passwd nagios
nagios:x:1001:1001:~/home/nagios:/bin/bash
voclabs:~ $ getent group nagios
nagios:x:1001:
voclabs:~ $ █
```

Rerun `sudo make install all`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install all
cd ./base && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.9/base'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagiosstats /usr/local/nagios/bin
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.9/base'
cd ./cgi && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.9/cgi'
make install-basic
make[2]: Entering directory '/tmp/nagioscore-nagios-4.4.9/cgi'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/sbin
for file in *.cgi; do \
    /usr/bin/install -c -s -m 775 -o nagios -g nagios $file /usr/local/nagios/sbin; \
done
make[2]: Leaving directory '/tmp/nagioscore-nagios-4.4.9/cgi'
make[1]: Leaving directory '/tmp/nagioscore-nagios-4.4.9/cgi'
cd ./html && make install
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.9/html'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/media
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/stylesheets
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/contexthelp
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/js
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images/logos
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/includes
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/ssi
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs/angular-1.3.9
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs/ui-utils-0.2.3
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/bootstrap-3.3.7
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/bootstrap-3.3.7/css
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/d3
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/spin
/usr/bin/install -c -m 664 -o nagios -g nagios robots.txt /usr/local/nagios/share
rm -f /usr/local/nagios/share/index.html
rm -f /usr/local/nagios/share/main.html
```

Install Init Script:

```
sudo make install-init
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-init
/usr/bin/install -c -m 755 -d -o root -g root /lib/systemd/system
/usr/bin/install -c -m 755 -o root -g root startup/default-service /lib/systemd/system/nagios.service
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Set Up Command Mode:

```
sudo make install-commandmode
```

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```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-commandmode
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***

voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Install Configuration Files:

`sudo make install-config`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-init
/usr/bin/install -c -m 755 -d -o root -g root /lib/systemd/system
/usr/bin/install -c -m 755 -o root -g root startup/default-service /lib/systemd/system/nagios.service
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-commandmode
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***

voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo make install-config
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/local/nagios/etc/nagios.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/local/nagios/etc/cgi.cfg
/usr/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /usr/local/nagios/etc/resource.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/templates.cfg /usr/local/nagios/etc/objects/templates.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/commands.cfg /usr/local/nagios/etc/objects/commands.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/contacts.cfg /usr/local/nagios/etc/objects/contacts.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/timeperiods.cfg /usr/local/nagios/etc/objects/timeperiods.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/localhost.cfg /usr/local/nagios/etc/objects/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/objects/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/objects/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/objects/switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Start Nagios: After completing the above steps, start Nagios with:

`sudo systemctl start nagios`

`sudo systemctl enable nagios`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl start nagios
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl enable nagios
Created symlink /etc/systemd/system/multi-user.target.wants/nagios.service → /usr/lib/systemd/system/nagios.service.
```

Check Status: To confirm that Nagios is running, you can check its status with:

```
sudo systemctl status nagios
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl start nagios
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl enable nagios
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl status nagios
• nagios.service - Nagios Core 4.4.9
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Wed 2024-10-23 08:12:32 UTC; 34s ago
     Docs: https://www.nagios.org/documentation
   Main PID: 58746 (nagios)
    Tasks: 6 (limit: 2255)
   Memory: 4.0M
      CPU: 29ms
   CGroup: /system.slice/nagios.service
           └─58746 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
           └─58747 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
           └─58748 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
           └─58749 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
           └─58750 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
           └─58755 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successfully initialized
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: qh: core query handler registered
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: qh: echo service query handler registered
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: qh: help for the query handler registered
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: wproc: Successfully registered manager as @wproc with query handler
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: wproc: Registry request: name=Core Worker 58747;pid=58747
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: wproc: Registry request: name=Core Worker 58750;pid=58750
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: wproc: Registry request: name=Core Worker 58749;pid=58749
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: wproc: Registry request: name=Core Worker 58748;pid=58748
Oct 23 08:12:32 ip-172-31-37-0.ec2.internal nagios[58746]: Successfully launched command file worker with pid 58755
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Installing Apache

Check for Available Package Managers

1. Use **yum** (if on CentOS or RHEL-based systems): If you're on a Red Hat-based system, you might use **yum** instead:

```
sudo yum install httpd
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo yum install httpd
Last metadata expiration check: 2:00:09 ago on Wed Oct 23 06:13:39 2024.
Package httpd-2.4.62-1.amzn2023.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Check Apache Status

Run the following command to check if Apache is running:

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`sudo systemctl status httpd`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl status httpd
o httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: inactive (dead)
     Docs: man:httpd.service(8)
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Start Apache

Run the following command to start the Apache server:

`sudo systemctl start httpd`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl start httpd
```

Check the Status Again

After starting it, check the status again to ensure it's running:

`sudo systemctl status httpd`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl start httpd
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Wed 2024-10-23 08:15:16 UTC; 11s ago
     Docs: man:httpd.service(8)
  Main PID: 59636 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"
    Tasks: 177 (limit: 2255)
   Memory: 17.8M
      CPU: 102ms
   CGroup: /system.slice/httpd.service
            └─59636 /usr/sbin/httpd -DFOREGROUND
              └─59638 /usr/sbin/httpd -DFOREGROUND
                └─59642 /usr/sbin/httpd -DFOREGROUND
                  └─59644 /usr/sbin/httpd -DFOREGROUND
                    └─59645 /usr/sbin/httpd -DFOREGROUND

Oct 23 08:15:15 ip-172-31-37-0.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 23 08:15:16 ip-172-31-37-0.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 23 08:15:16 ip-172-31-37-0.ec2.internal httpd[59636]: Server configured, listening on: port 80
voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

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Enable Apache to Start on Boot

If it's running, enable it to start on boot with:

```
sudo systemctl enable httpd
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
```

Access Nagios

Now try accessing the Nagios web interface again:

```
curl -I http://localhost/nagios
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
voclabs:/tmp/nagioscore-nagios-4.4.9 $ curl -I http://localhost/nagios
HTTP/1.1 401 Unauthorized
Date: Wed, 23 Oct 2024 08:16:24 GMT
Server: Apache/2.4.62 (Amazon Linux)
WWW-Authenticate: Basic realm="Nagios Access"
Content-Type: text/html; charset=iso-8859-1

voclabs:/tmp/nagioscore-nagios-4.4.9 $ █
```

Check on browser <http://3.94.99.227/nagios/>

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← → ↻ 34.238.163.130/nagios/ ☆ 🔊 ⋮

Sign in

http://34.238.163.130
Your connection to this site is not private

Username

Password

Sign in Cancel

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Nagios® Core™

✓ Daemon running with PID 58746

Nagios® Core™
Version 4.4.9
November 16, 2022
[Check for updates](#)

A new version of Nagios Core is available!
[Visit nagios.org to download Nagios 4.5.6.](#)

Get Started

- Start monitoring your infrastructure
- Change the look and feel of Nagios
- Extend Nagios with hundreds of addons
- Get support
- Get training
- Get certified

Quick Links

- Nagios Library (tutorials and docs)
- Nagios Labs (development blog)
- Nagios Exchange (plugins and addons)
- Nagios Support (tech support)
- Nagios.com (company)
- Nagios.org (project)

Latest News

Don't Miss...

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Login to Nagios

- Use the credentials you set up earlier. If you followed the previous steps, the username is likely **nagiosadmin**.
- You will be prompted for a password, which you set with:
bash

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Copy code

```
sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
```

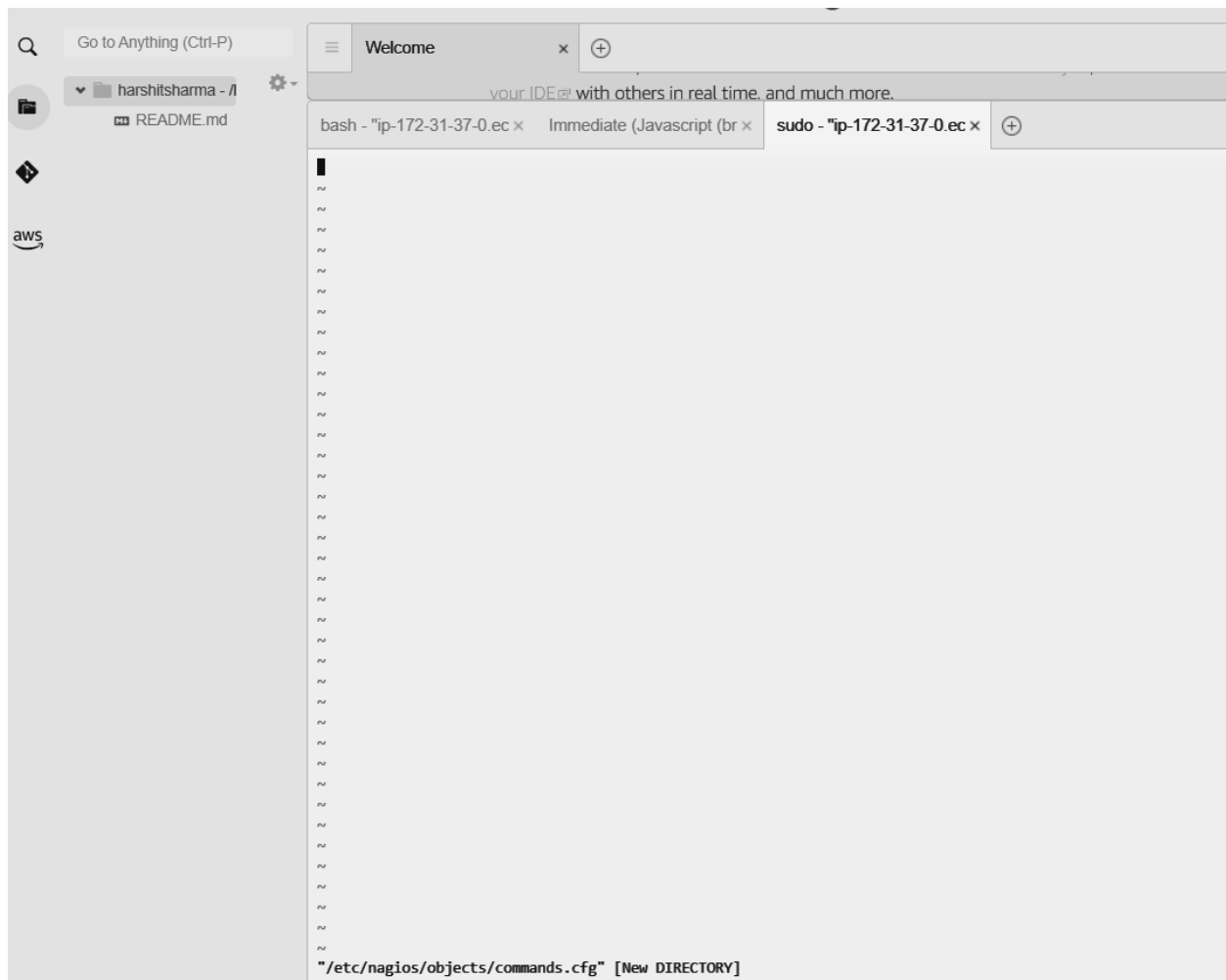
```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
voclabs:/tmp/nagioscore-nagios-4.4.9 $
```

Password:harshit006@

Define a Nagios Command

1. Open the Commands Configuration File:

```
sudo vim /etc/nagios/objects/commands.cfg
```



Add the Command: Add the following lines to define the command to check the Nginx pod's availability. You may want to ensure the command is added within the existing `define command` blocks or at the end of the file:

```
define command {  
  
    command_name      check_nginx_pod  
  
    command_line      /usr/local/bin/kubectl get pods | grep nginx |  
grep Running  
  
}
```

```
service {  
    use                generic-service  
    host_name          localhost  
    service_description Check Nginx Pod  
    check_command       check_nginx_pod  
}
```

Restart Nagios to apply the changes:

```
sudo systemctl restart nagios
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ sudo systemctl restart nagios
```

Check the status of Nagios to confirm it's running:

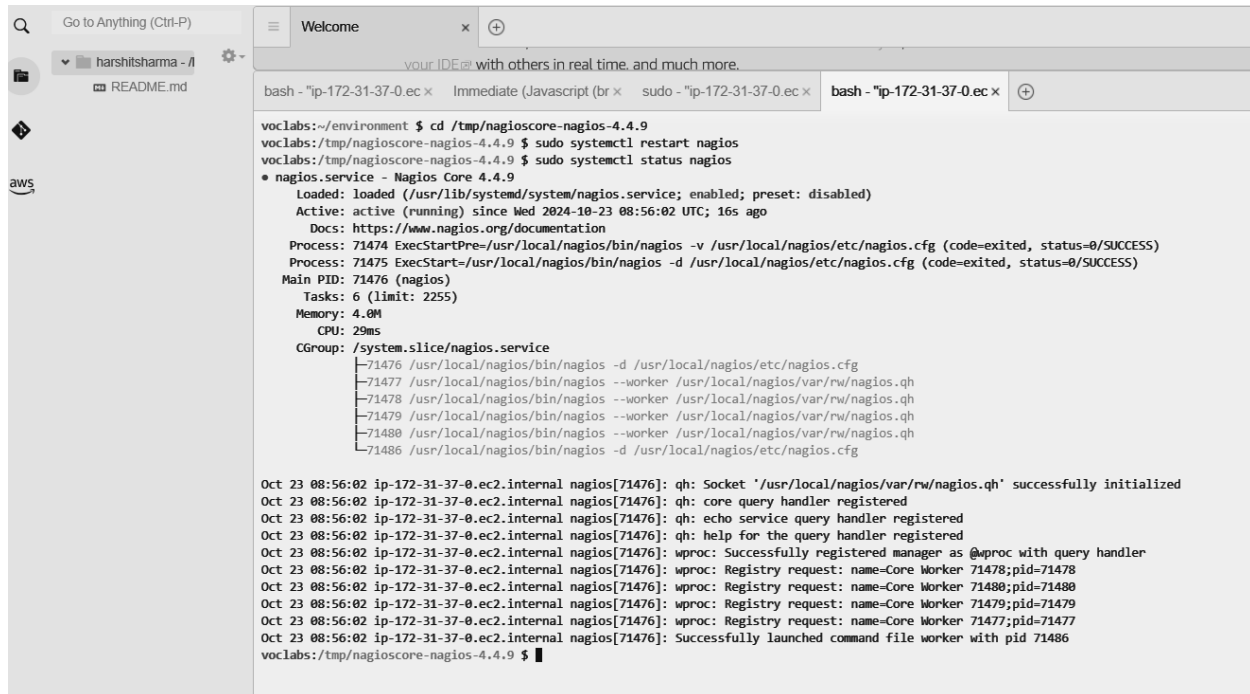
```
sudo systemctl status nagios
```

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The screenshot shows a terminal window with a 'Welcome' tab. The user is in a directory named 'voclabs' and has navigated to '/tmp/nagioscore-nagios-4.4.9'. They have executed the following commands:
1. `sudo systemctl restart nagios`
2. `sudo systemctl status nagios`
The output of the status command shows that the 'nagios.service' is loaded and active (running). It provides details about the service's configuration, including the path to the Nagios core and the configuration file. The output also shows the process details for the Nagios daemon and its workers, indicating that they have been successfully launched.

Changes:(not happening)

Verify That Nagios Can Detect When the Nginx Pod is Running or Not

1. Simulate Failure (Stop Nginx Pod):

1. Scale Down the Nginx Deployment to simulate a failure:

```
kubectl scale deployment nginx --replicas=0
```

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ kubectl scale deployment nginx --replicas=0
deployment.apps/nginx scaled
voclabs:/tmp/nagioscore-nagios-4.4.9 $
```

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Check Nagios: Open the Nagios web interface. You should see an alert indicating that the Nginx pod is down.

The screenshot shows the Nagios web interface at 34.238.163.130/nagios/. The interface includes a sidebar with navigation links like General, Home, Documentation, Current Status, Tactical Overview, Map (Legacy), Hosts, Services, Host Groups, Summary, Grid, Service Groups, Summary, Grid, Problems, Services (Unhandled), Hosts (Unhandled), Network Outages, and Quick Search. The main content area displays 'Current Network Status' with a last update of Wed Oct 23 08:56:59 UTC 2024. It also shows 'Host Status Totals' and 'Service Status Totals'. A table titled 'Host Status Details For All Host Groups' shows one host, 'localhost', with a status of 'DOWN'. The status information for localhost indicates a failure: '(No output on stdout) stderr: execvp(/usr/local/nagios/libexec/check_ping...) failed. error is 2: No such file or directory'.

To Change the status to UP

Install Nagios Plugins

If the plugins are missing, follow these steps to install them:

Download Nagios Plugins:

```
cd /tmp
```

1. `curl -O https://nagios-plugins.org/download/nagios-plugins-2.4.3.tar.gz`

```
voclabs:/tmp/nagioscore-nagios-4.4.9 $ kubectl scale deployment nginx --replicas=0
deployment.apps/nginx scaled
voclabs:/tmp/nagioscore-nagios-4.4.9 $ cd ~
voclabs:~ $ cd /tmp
voclabs:/tmp $ curl -O https://nagios-plugins.org/download/nagios-plugins-2.4.3.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 2683k  100 2683k    0     0  6964k      0  --:--:-- --:--:-- --:--:--  6988k
voclabs:/tmp $
```

Extract the Plugins:

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```
tar -xzf nagios-plugins-2.4.3.tar.gz
```

```
cd nagios-plugins-2.4.3
```

```
voclabs:/tmp $ tar -xzf nagios-plugins-2.4.3.tar.gz
voclabs:/tmp $ cd nagios-plugins-2.4.3
```

Compile and Install the Plugins:

```
./configure --with-nagios-user=nagios --with-nagios-group=nagios
```

```
    --with-mysql: no
    --with-openssl: yes
    --with-gnutls: no
    --enable-extra-opts: yes
    --with-perl: /usr/bin/perl
    --enable-perl-modules: no
    --with-cgiurl: /nagios/cgi-bin
    --with-trusted-path: /usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
    --enable-libtap: no
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating gl/Makefile
config.status: creating nagios-plugins.spec
config.status: creating tools/build_perl_modules
config.status: creating Makefile
config.status: creating tap/Makefile
config.status: creating lib/Makefile
config.status: creating plugins/Makefile
config.status: creating lib/tests/Makefile
config.status: creating plugins-root/Makefile
config.status: creating plugins-scripts/Makefile
config.status: creating plugins-scripts/utils.pm
config.status: creating plugins-scripts/utils.sh
config.status: creating perlmods/Makefile
config.status: creating test.pl
config.status: creating pkg/solaris/pkginfo
config.status: creating po/Makefile.in
config.status: creating config.h
config.status: config.h is unchanged
config.status: executing depfiles commands
config.status: executing libtool commands
config.status: executing po-directories commands
config.status: creating po/POTFILES
config.status: creating po/Makefile
voclabs:/tmp/nagios-plugins-2.4.3 $
```

```
make
```

```

../gl/gl_openssl.h: In function 'sha1_finish_ctx':
../gl/gl_openssl.h:94:1: warning: 'SHA1_Final' is deprecated: Since OpenSSL 3.0 [-Wdeprecated-declarations]
 94 | { OPENSSL_FN (_Final) ((unsigned char *) res, (_GL_CTX *) ctx); return res; }
    | ^
In file included from /usr/include/openssl/x509.h:41,
                 from ../plugins/common.h:139,
                 from check_icmp.c:52:
/usr/include/openssl/sha.h:51:27: note: declared here
 51 | OSSL_DEPRECATEDIN_3_0 int SHA1_Final(unsigned char *md, SHA_CTX *c);
    | ^
In file included from ../gl/sha1.h:37,
                 from ../lib/utills_base.h:5,
                 from ../plugins/utills.h:17,
                 from ../plugins/netutils.h:35,
                 from check_icmp.c:53:
../gl/gl_openssl.h: In function 'sha1_read_ctx':
../gl/gl_openssl.h:105:3: warning: 'SHA1_Final' is deprecated: Since OpenSSL 3.0 [-Wdeprecated-declarations]
 105 | OPENSSL_FN (_Final) ((unsigned char *) res, &tmp_ctx);
     | ^
In file included from /usr/include/openssl/x509.h:41,
                 from ../plugins/common.h:139,
                 from check_icmp.c:52:
/usr/include/openssl/sha.h:51:27: note: declared here
 51 | OSSL_DEPRECATEDIN_3_0 int SHA1_Final(unsigned char *md, SHA_CTX *c);
    | ^
mv -f .deps/check_icmp.Tpo .deps/check_icmp.Po
/bin/sh ../libtool --tag=CC --mode=link gcc -DNP_VERSION="2.4.3" -g -O2 -L -o check_icmp check_icmp.o ../plugins/netutils.o ../plugins/utills.o ../lib/libnagiosplug.a ../gl/libgnu.a -lresolv -lssl -lcrypto -ldl
libtool: link: gcc -DNP_VERSION="2.4.3" -g -O2 -o check_icmp check_icmp.o ../plugins/netutils.o ../plugins/utills.o -L. -L../lib/libnagiosplug.a -L../gl/libgnu.a -lresolv -lssl -lcrypto -ldl
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-root'
Making all in po
make[2]: Entering directory '/tmp/nagios-plugins-2.4.3/po'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3/po'
make[2]: Entering directory '/tmp/nagios-plugins-2.4.3/po'
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3/po'
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3'
voclabs:/tmp/nagios-plugins-2.4.3 $ █

```

sudo make install

```

make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-scripts'
Making install in plugins-root
make[1]: Entering directory '/tmp/nagios-plugins-2.4.3/plugins-root'
make[2]: Entering directory '/tmp/nagios-plugins-2.4.3/plugins-root'
/usr/bin/install -c -o nagios -g nagios check_dhcp /usr/local/nagios/libexec/check_dhcp
chown root /usr/local/nagios/libexec/check_dhcp
chmod ug=rx,u+s /usr/local/nagios/libexec/check_dhcp
/usr/bin/install -c -o nagios -g nagios check_icmp /usr/local/nagios/libexec/check_icmp
chown root /usr/local/nagios/libexec/check_icmp
chmod ug=rx,u+s /usr/local/nagios/libexec/check_icmp
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-root'
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-root'
Making install in po
make[1]: Entering directory '/tmp/nagios-plugins-2.4.3/po'
/usr/bin/mkdir -p /usr/local/nagios/share
installing fr.gmo as /usr/local/nagios/share/locale/fr/LC_MESSAGES/nagios-plugins.mo
installing de.gmo as /usr/local/nagios/share/locale/de/LC_MESSAGES/nagios-plugins.mo
if test "nagios-plugins" = "gettext-tools"; then \
  /usr/bin/mkdir -p /usr/local/nagios/share/gettext/po; \
  for file in Makefile.in.in remove-potcdate.sin Makevars.template; do \
    /usr/bin/install -c -o nagios -g nagios -m 644 ./file \
      /usr/local/nagios/share/gettext/po/$file; \
  done; \
  for file in Makevars; do \
    rm -f /usr/local/nagios/share/gettext/po/$file; \
  done; \
else \
  : ; \
fi
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3/po'
make[1]: Entering directory '/tmp/nagios-plugins-2.4.3'
make[2]: Entering directory '/tmp/nagios-plugins-2.4.3'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3'
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3'
voclabs:/tmp/nagios-plugins-2.4.3 $ █

```

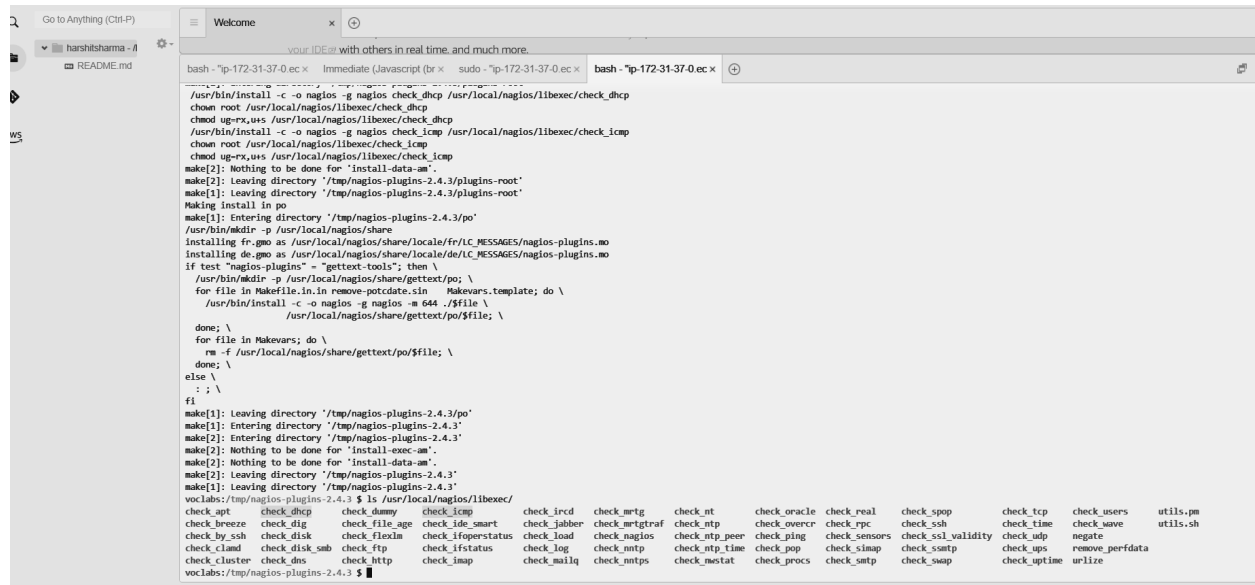
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ls /usr/local/nagios/libexec/



```
bash - "ip-172-31-37-0.ec x" sudo "ip-172-31-37-0.ec x" bash - "ip-172-31-37-0.ec x"
/usr/bin/install -c -o nagios -g nagios check_dhcp /usr/local/nagios/libexec/check_dhcp
chmod up-rx,ws /usr/local/nagios/libexec/check_dhcp
/usr/bin/install -c -o nagios -g nagios check_icmp /usr/local/nagios/libexec/check_icmp
chmod up-rx,ws /usr/local/nagios/libexec/check_icmp
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-root'
make[3]: Leaving directory '/tmp/nagios-plugins-2.4.3/plugins-root'
Making install in po
make[1]: Entering directory '/tmp/nagios-plugins-2.4.3/po'
/usr/bin/mkdir -p /usr/local/nagios/share
Installing fr.gmo as /usr/local/nagios/share/locale/fr/LC_MESSAGES/nagios-plugins.mo
Installing de.gmo as /usr/local/nagios/share/locale/de/LC_MESSAGES/nagios-plugins.mo
if test "nagios-plugins" = "gettext-tools"; then \
  /usr/bin/mkdir -p /usr/local/nagios/share/gettext/po; \
  for file in Makefile.in.in remove-potdate.sin Makevars.template; do \
    /usr/bin/install -c -o nagios -g nagios -m 644 ./$$file \
      /usr/local/nagios/share/gettext/po/$$file; \
  done; \
  for file in Makevars; do \
    rm -f /usr/local/nagios/share/gettext/po/$$file; \
  done; \
else \
  ; \
fi
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3/po'
make[1]: Entering directory '/tmp/nagios-plugins-2.4.3'
make[2]: Entering directory '/tmp/nagios-plugins-2.4.3'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-2.4.3'
make[1]: Leaving directory '/tmp/nagios-plugins-2.4.3'
voclabs:/tmp/nagios-plugins-2.4.3 $ ls /usr/local/nagios/libexec/
check_apt      check_dhcp      check_dummy     check_icmp      check_ircd      check_mrtg      check_nt        check_oracle    check_real      check_spop      check_tcp      check_users     utils.pm
check_breeze   check_dig        check_file_age  check_ide_smart  check_ircd      check_mrtgtraf  check_ntp       check_overcr    check_rpc       check_ssh       check_time     check_wave      utils.sh
check_by_ssh   check_disk       check_filein    check_ifoperstatus  check_load      check_nagios    check_ntp_peer  check_ping      check_sensors  check_ssl_validity  check_udp     negate
check_cband    check_disk_smb   checkftp        check_ifstatus    check_log        check_nttp      check_ntp_time  check_pop       check_smap     check_sntp     check_ups      remove_perdata
check_cluster  check_dns        check_http      check_inap       check_mailq      check_nttps     check_nstata    check_procs     check_sntp     check_smap     check_uptime   urlize
voclabs:/tmp/nagios-plugins-2.4.3 $
```

sudo systemctl restart nagios

```
voclabs:/tmp/nagios-plugins-2.4.3 $ sudo systemctl restart nagios
voclabs:/tmp/nagios-plugins-2.4.3 $
```

Now Status is UP

←→↻

Not secure 34.238.163.130/nagios/

☆

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Current Network Status

Last Updated: Wed Oct 23 09:14:31 UTC 2024

Updated every 90 seconds

Nagios® Core™ 4.4.9 - www.nagios.org

Logged in as nagiosadmin

View Service Status Detail For All Host Groups

View Status Overview For All Host Groups

View Status Summary For All Host Groups

View Status Grid For All Host Groups

Host Status Totals

Up Down Unreachable Pending

1 0 0 0

All Problems All Types

0 1

Service Status Totals

Ok Warning Unknown Critical Pending

6 1 0 1 0

All Problems All Types

2 8

Host Status Details For All Host Groups

Limit Results: 100

Host

localhost

Status

UP

Last Check

10-23-2024 09:13:21

Duration

0d 0h 4m 29s

Status Information

PING OK - Packet loss = 0%, RTA = 0.06 ms

Results 1 - 1 of 1 Matching Hosts

Current Network Status

Last Updated: Wed Oct 23 09:14:31 UTC 2024

Updated every 90 seconds

Nagios® Core™ 4.4.9 - www.nagios.org

Logged in as nagiosadmin

View Service Status Detail For All Host Groups

View Status Overview For All Host Groups

View Status Summary For All Host Groups

View Status Grid For All Host Groups

Host Status Totals

Up Down Unreachable Pending

1 0 0 0

All Problems All Types

0 1

Service Status Totals

Ok Warning Unknown Critical Pending

6 1 0 1 0

All Problems All Types

2 8

Host Status Details For All Host Groups

Limit Results: 100

Host

localhost

Status

UP

Last Check

10-23-2024 09:13:21

Duration

0d 0h 4m 29s

Status Information

PING OK - Packet loss = 0%, RTA = 0.06 ms

Results 1 - 1 of 1 Matching Hosts

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Last Updated: Wed Oct 23 09:15:44 UTC 2024

Updated every 90 seconds

Nagios® Core™ 4.4.9 - www.nagios.org

Logged in as nagiosadmin

View Status Detail For This Host

View Alert History For This Host

View Availability Report For This Host

View Notifications For This Host

Host

localhost

(localhost)

Member of

linux-servers

127.0.0.1

Host State Information

Host Status:

UP (for 0d 0h 5m 42s)

Status Information:

PING OK - Packet loss = 0%, RTA = 0.06 ms

Performance Data:

rta=0.059000ms;3000.000000;5000.000000;0.000000 pl=0%;80,100,0

Current Attempt:

1/10 (HARD state)

Last Check Time:

10-23-2024 09:13:21

Check Type:

ACTIVE

Check Latency / Duration:

0.000 / 4.173 seconds

Next Scheduled Active Check:

10-23-2024 09:18:21

Last State Change:

10-23-2024 09:10:02

Last Notification:

10-23-2024 09:10:06 (notification 0)

Is This Host Flapping?

NO (5.07% state change)

Is Scheduled Downtime?

NO

Last Update:

10-23-2024 09:15:35 (0d 0h 0m 9s ago)

Active Checks:

ENABLED

Passive Checks:

ENABLED

Obsessing:

ENABLED

Notifications:

ENABLED

Event Handler:

ENABLED

Flap Detection:

ENABLED

Host Commands

Disable active checks of this host

Re-schedule the next check of this host

Submit passive check result for this host

Stop accepting passive checks for this host

Stop obsessing over this host

Disable notifications for this host

Send custom host notification

Schedule downtime for this host

Schedule downtime for all services on this host

Disable notifications for all services on this host

Enable notifications for all services on this host

Schedule a check of all services on this host

Disable checks of all services on this host

Enable checks of all services on this host

Disable event handler for this host

Disable flap detection for this host

Clear flapping state for this host

Host Comments

Add a new comment

Delete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This host has no comments associated with it

Conclusion:

Name:HARSHIT SHARMA

19/10/2024

Div:D15B

Roll No.:55

CONCLUSION::This project not only demonstrates the practical application of deploying a simple application using Kubernetes but also highlights the importance of monitoring to ensure system reliability and performance. By combining AWS Cloud9, Kubernetes, Nginx, and Nagios, developers can create a robust and easily manageable application architecture.